

```
In [3]: from sklearn.linear_model import SGDRegressor
from sklearn.multioutput import MultiOutputRegressor
import numpy as np
import matplotlib.pyplot as plt

X = np.array([
    [1, 2],
    [2, 1],
    [3, 4],
    [4, 3],
    [5, 5],
    [6, 7],
    [7, 6]
])

Y = np.array([
    [5, 8],
    [6, 9],
    [9, 12],
    [10, 13],
    [13, 16],
    [16, 20],
    [17, 21]
])

sgd = SGDRegressor(
    max_iter=1000,
    eta0=0.01,
    learning_rate='constant',
    random_state=42
)

model = MultiOutputRegressor(sgd)

model.fit(X, Y)

Y_pred = model.predict(X)

print("\nActual Outputs")
print(Y)

print("\nPredicted Outputs")
print(np.round(Y_pred, 2))

new_sample = np.array([[8, 7]])
prediction = model.predict(new_sample)

print("\nPrediction for", new_sample)
print(prediction)
```

```
plt.figure(figsize=(6,4))
plt.scatter(Y[:,0], Y_pred[:,0], color='blue')
plt.plot([Y[:,0].min(), Y[:,0].max()],
         [Y[:,0].min(), Y[:,0].max()],
         'r--')
plt.xlabel("Actual Output 1")
plt.ylabel("Predicted Output 1")
plt.title("Output 1: Actual vs Predicted")
plt.grid(True)
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(Y[:,1], Y_pred[:,1], color='green')
plt.plot([Y[:,1].min(), Y[:,1].max()],
         [Y[:,1].min(), Y[:,1].max()],
         'r--')
plt.xlabel("Actual Output 2")
plt.ylabel("Predicted Output 2")
plt.title("Output 2: Actual vs Predicted")
plt.grid(True)
plt.show()
```

Actual Outputs

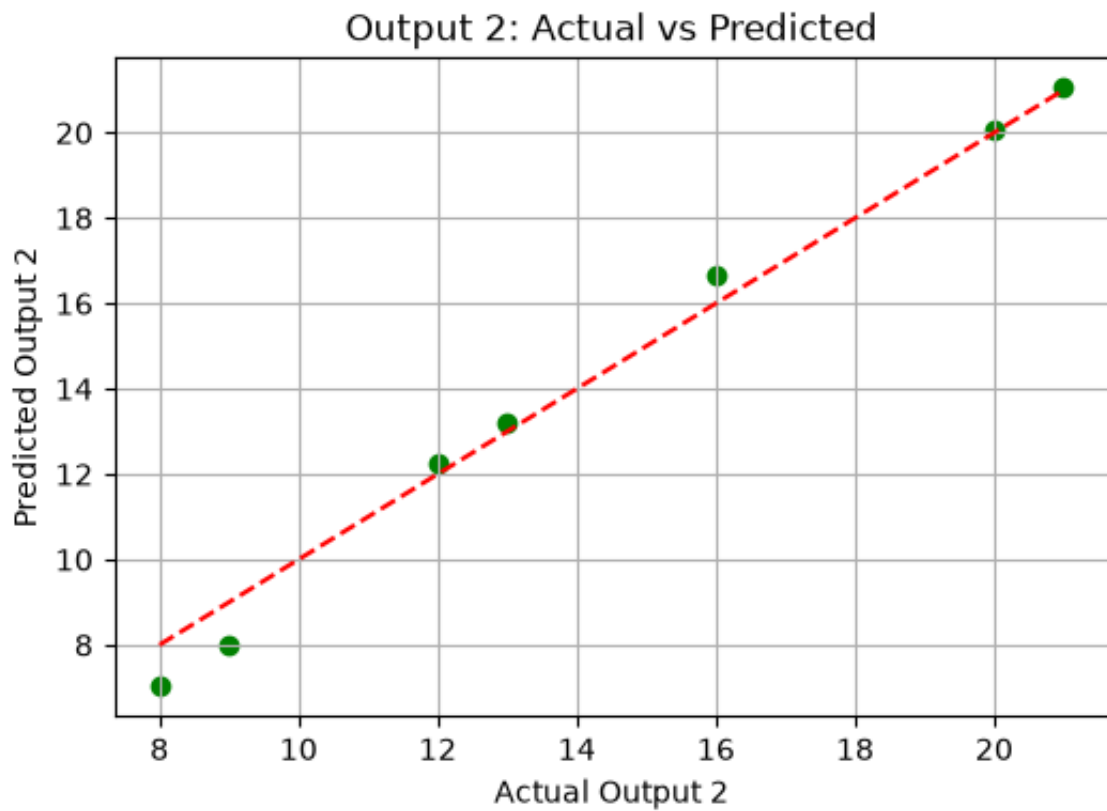
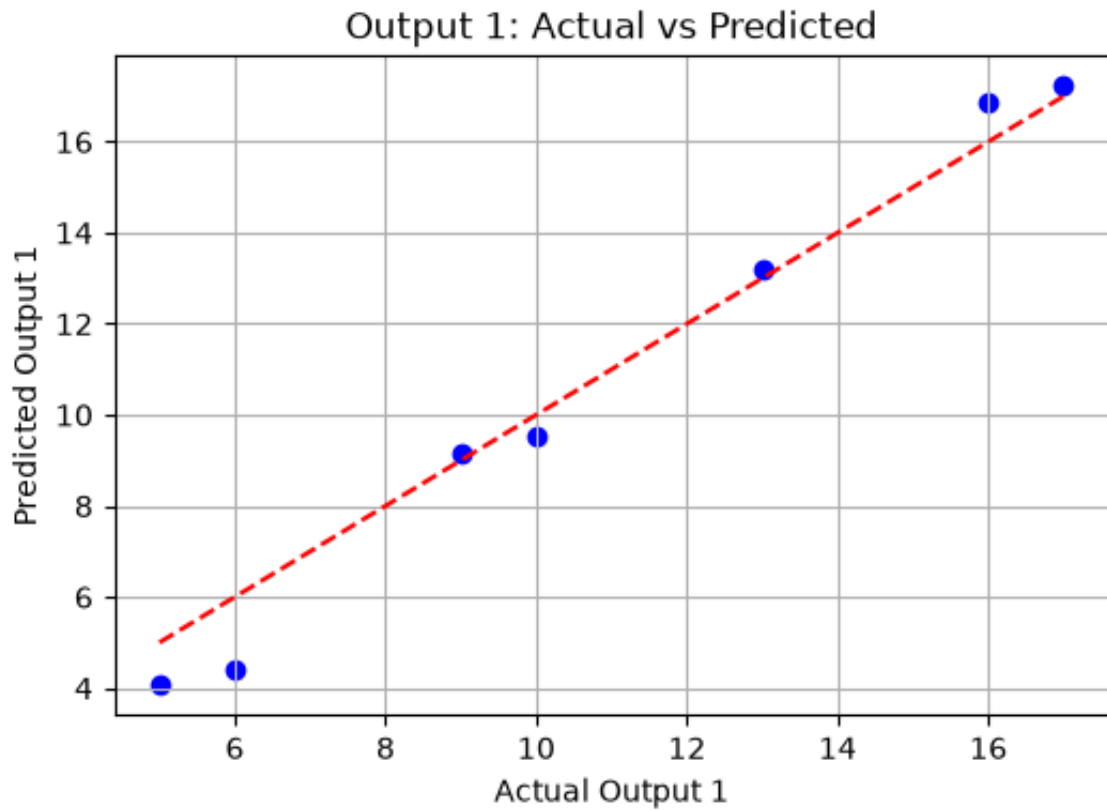
```
[[ 5  8]
 [ 6  9]
 [ 9 12]
 [10 13]
 [13 16]
 [16 20]
 [17 21]]
```

Predicted Outputs

```
[[ 4.05  7.02]
 [ 4.41  8.  ]
 [ 9.18 12.24]
 [ 9.53 13.22]
 [13.2  16.64]
 [16.87 20.06]
 [17.22 21.04]]
```

Prediction for [[8 7]]

```
[[19.78065588 23.65065574]]
```



In []: